Lesson 04 Demo 05

Implementing Protected Routes in a React Application

Objective: To implement protected routes in a Vite-based React app for secure navigation based on user authentication

Tools required: Visual Studio Code and Node.js

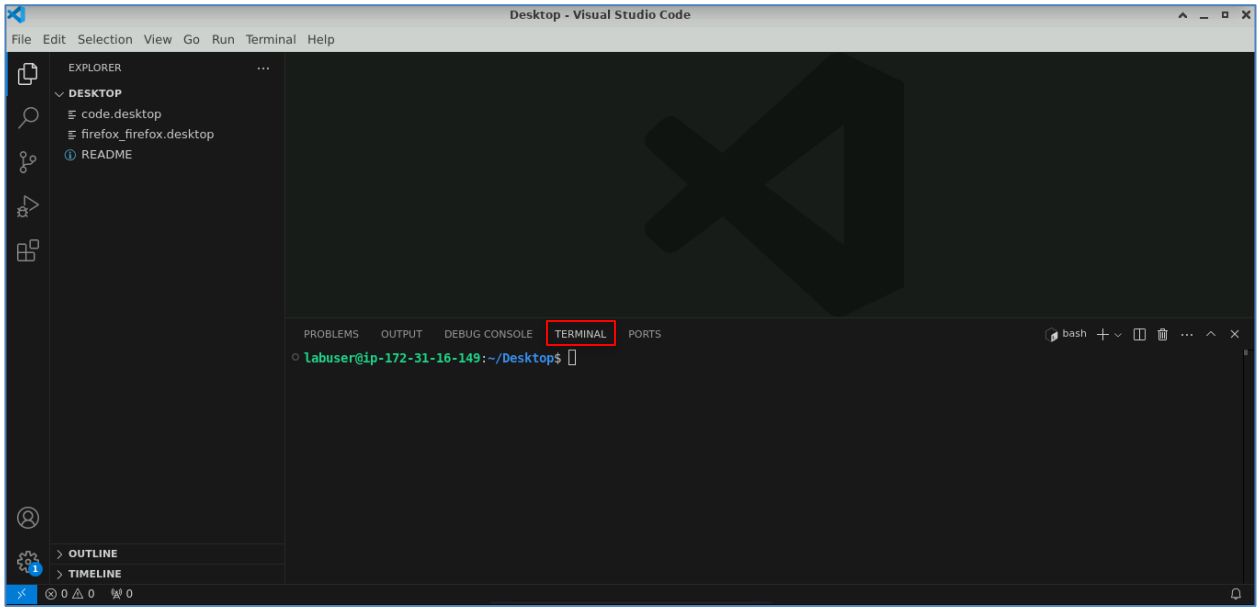
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Create React components
3. Import and use the components
4. Run and verify the application

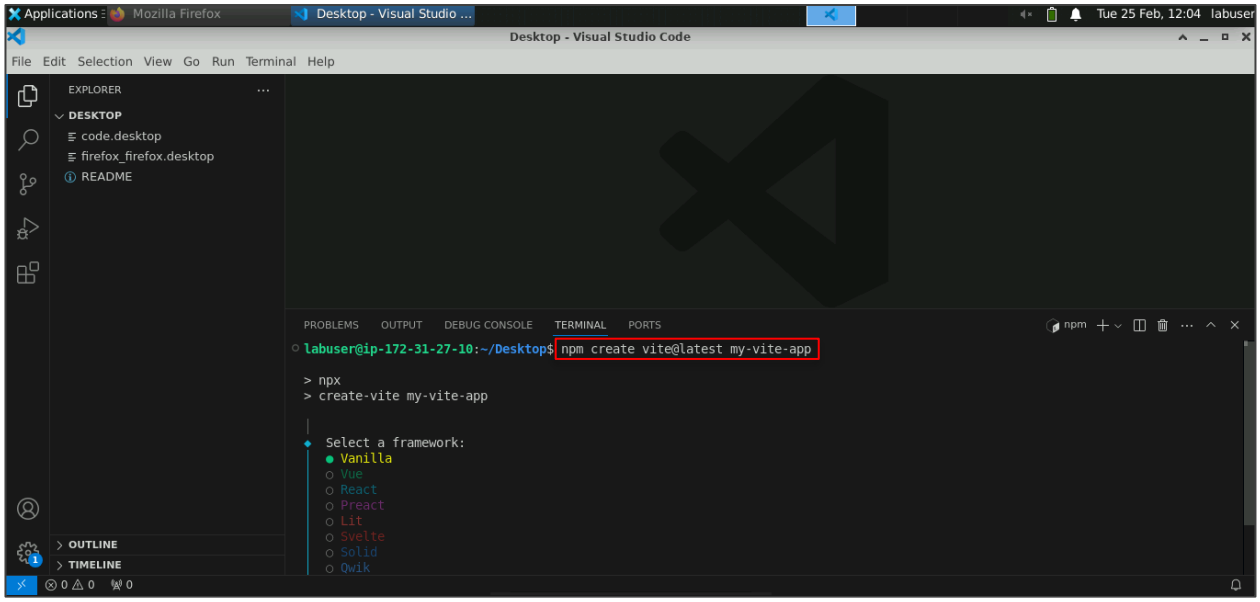
Step 1: Set up a new React project using Vite

1.1 Navigate to VS Code and open the terminal

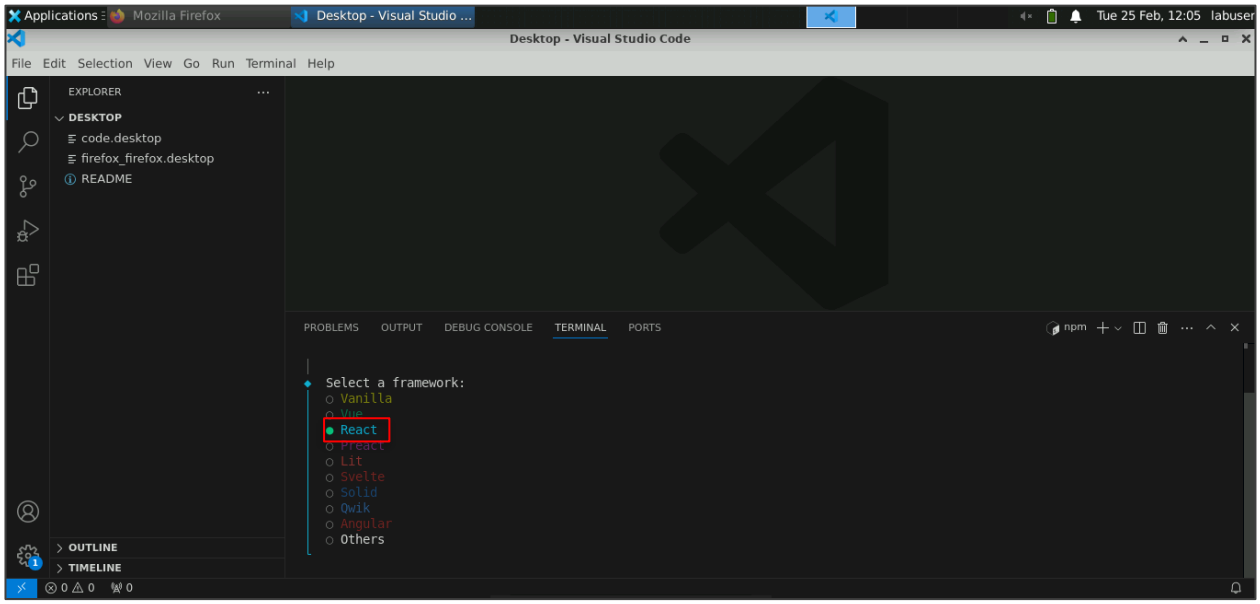


1.2 Run the following command to create a new React project using Vite:

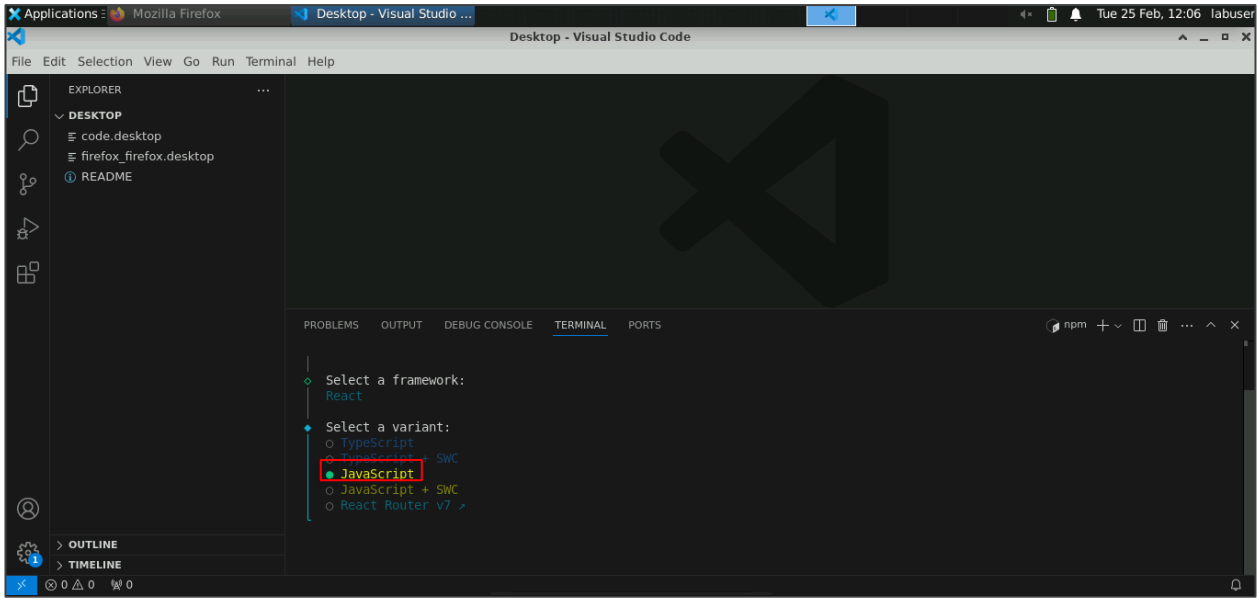
npm create vite@latest my-vite-app



1.3 Select **React** and press enter

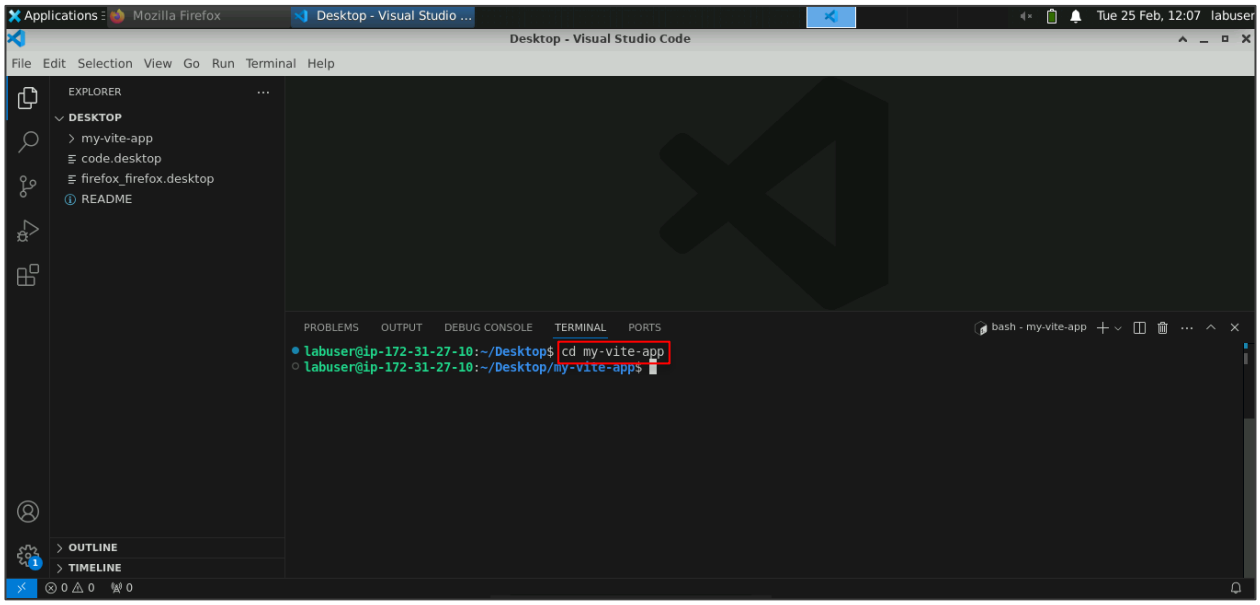


1.4 Select **JavaScript** and press enter



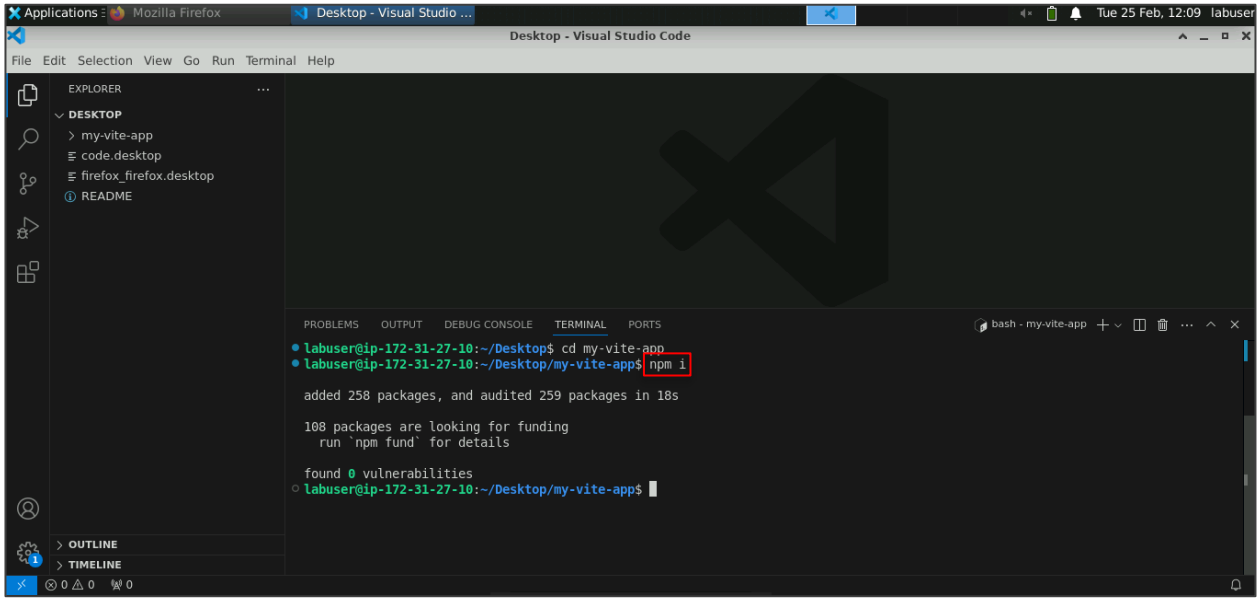
1.5 Run the following command to navigate to the project folder:

cd my-vite-app



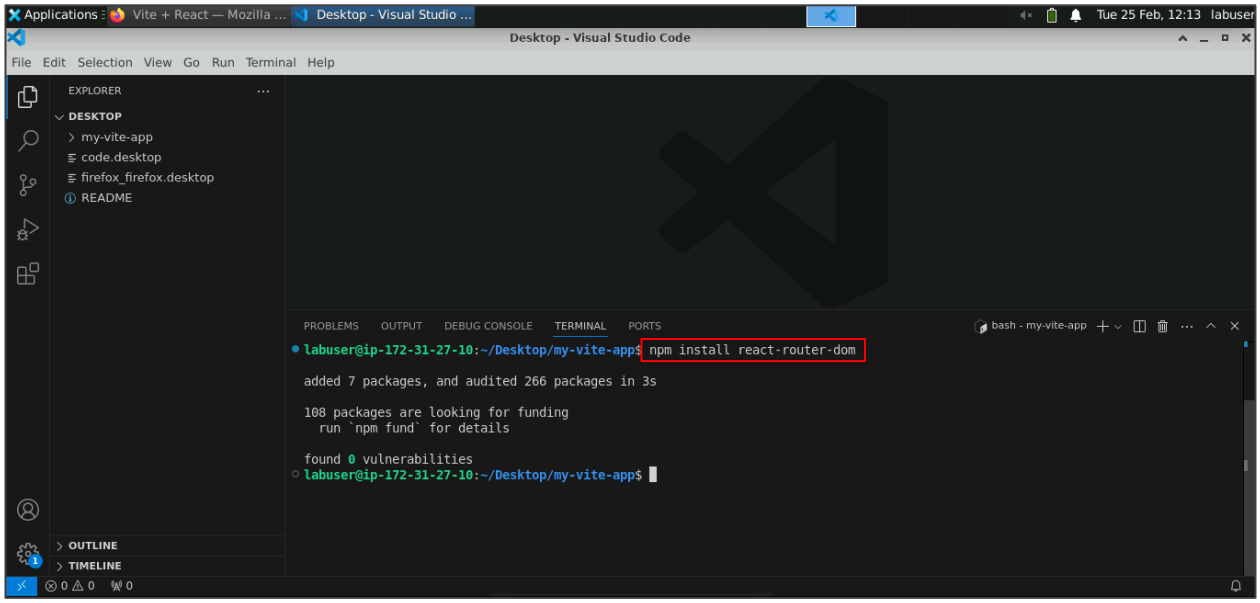
1.6 Run the following command to install dependencies:

npm i



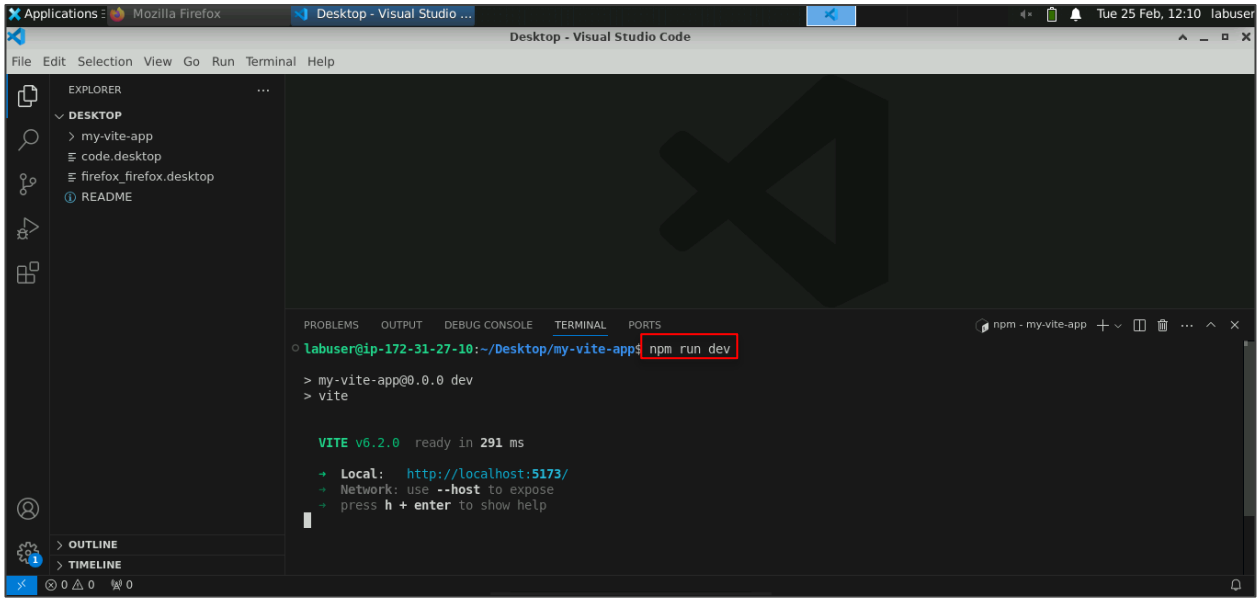
1.7 Run the following command to install the React router:

npm install react-router-dom



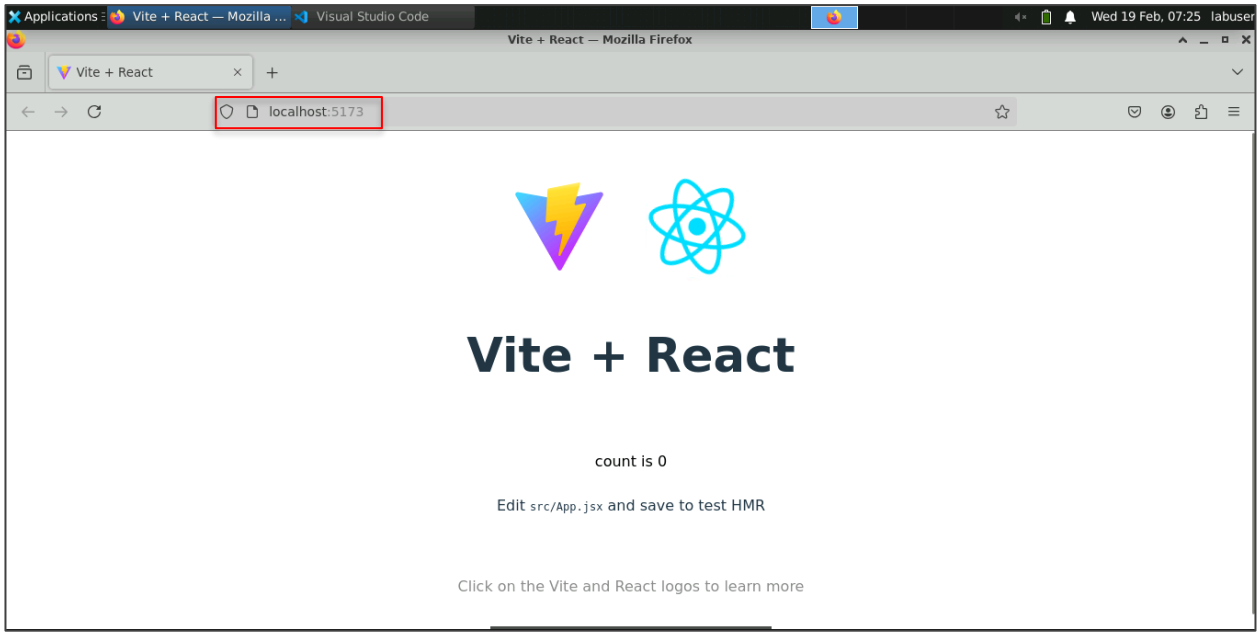
1.8 Run the following command to start the development server:

npm run dev



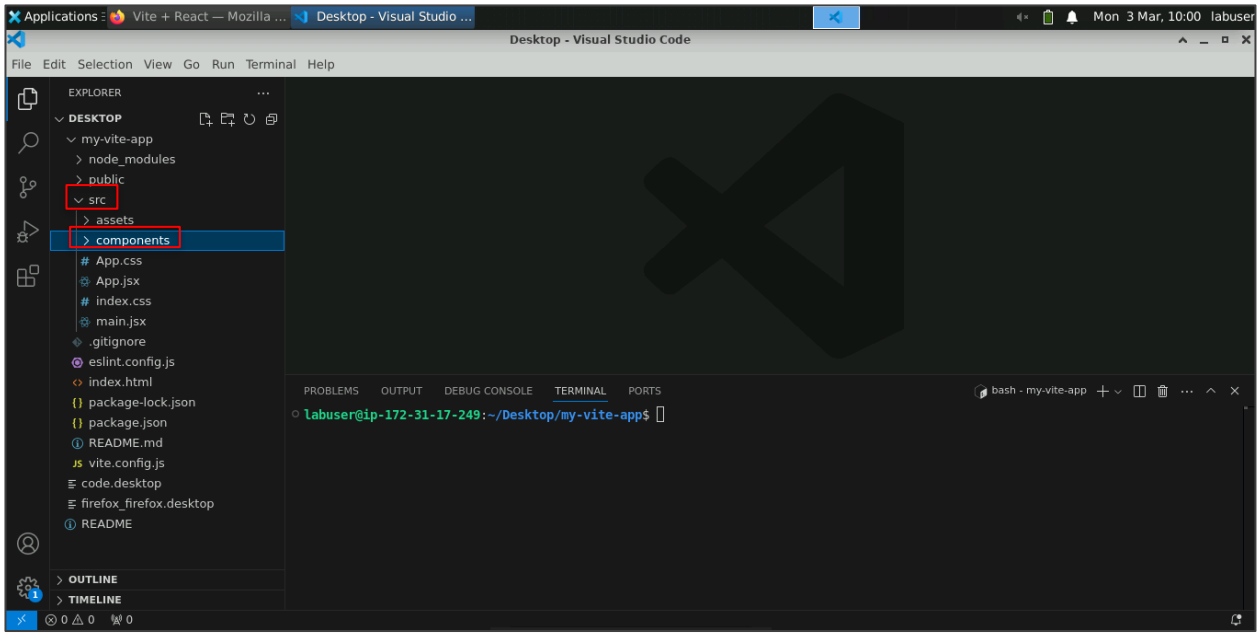
1.9 Navigate to the web browser and paste the link below:

http://localhost:5173

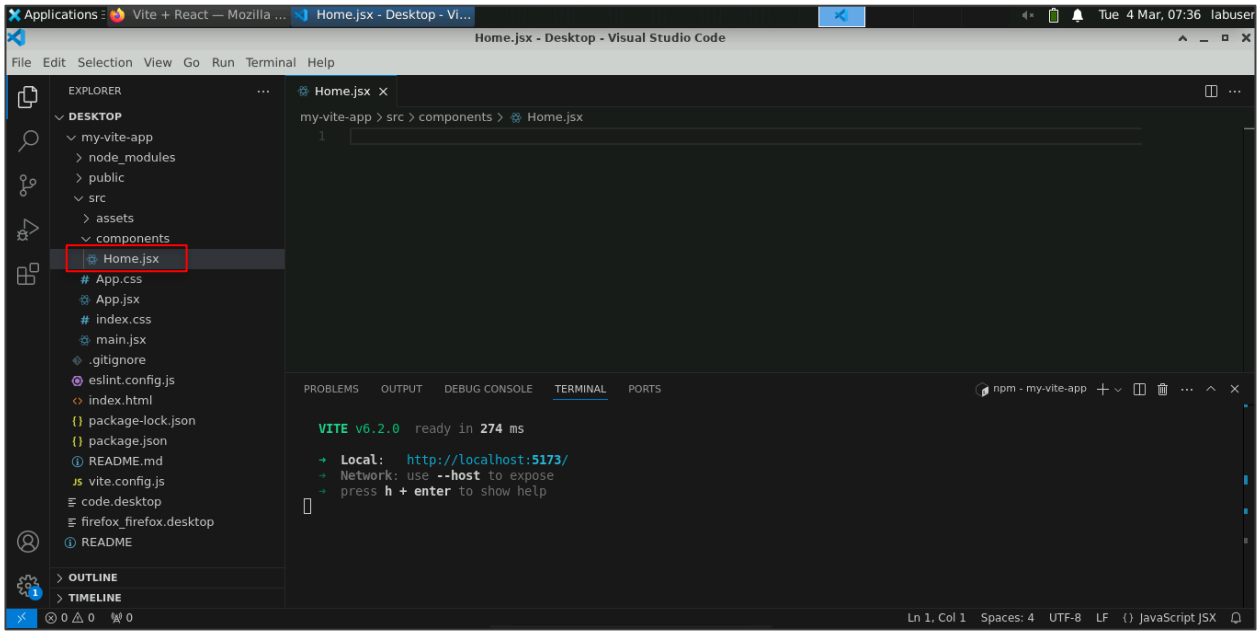


Step 2: Create React components

2.1 Inside the **src** folder, create a new folder named **components**

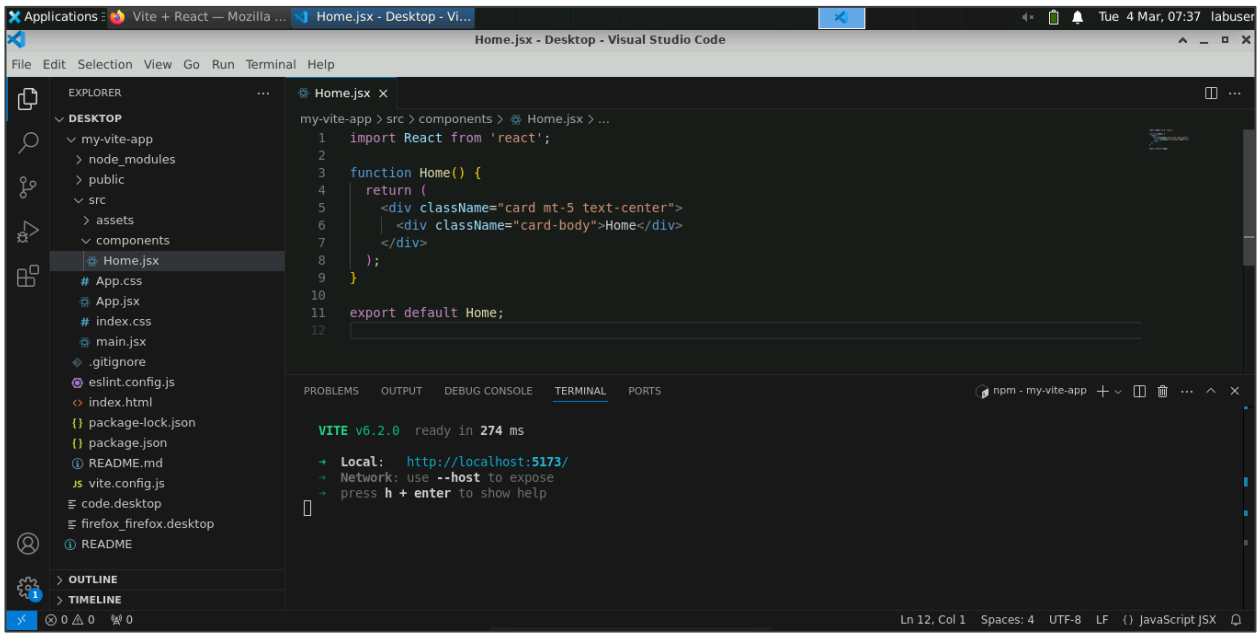


2.2 Inside the **components** folder, create a new file named **Home.jsx**

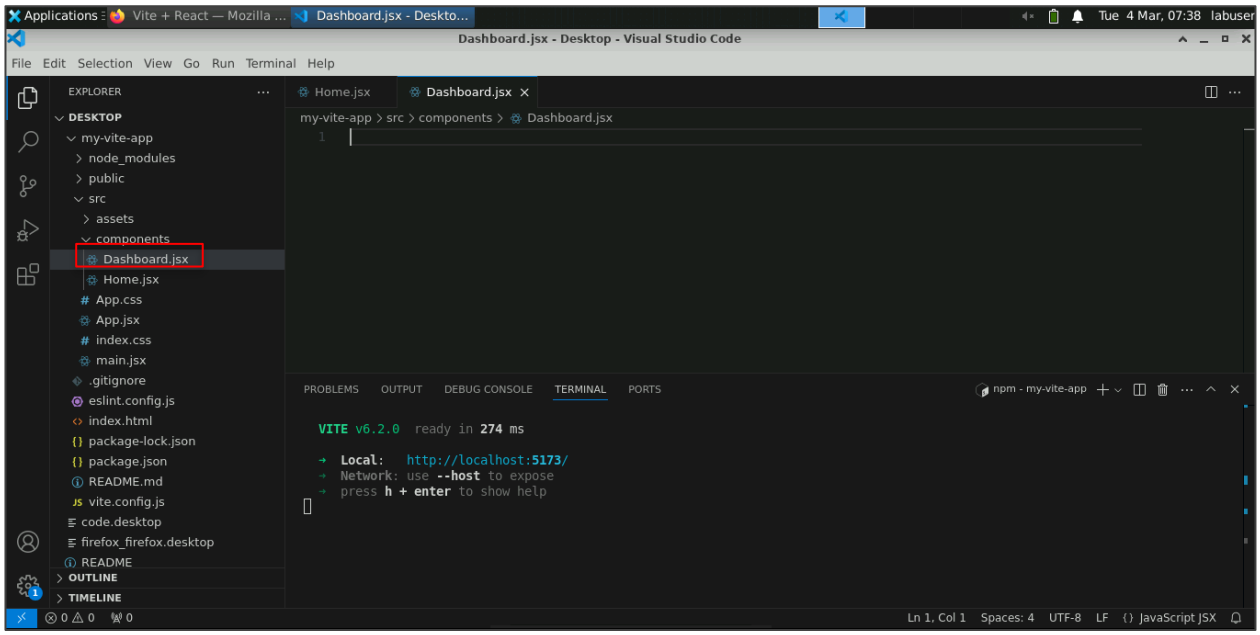


2.3 Add the following code to **Home.jsx:**

```
import React from 'react';

function Home() {
  return (
    <div className="card mt-5 text-center">
      <div className="card-body">Home</div>
    </div>
  );
}

export default Home;
```

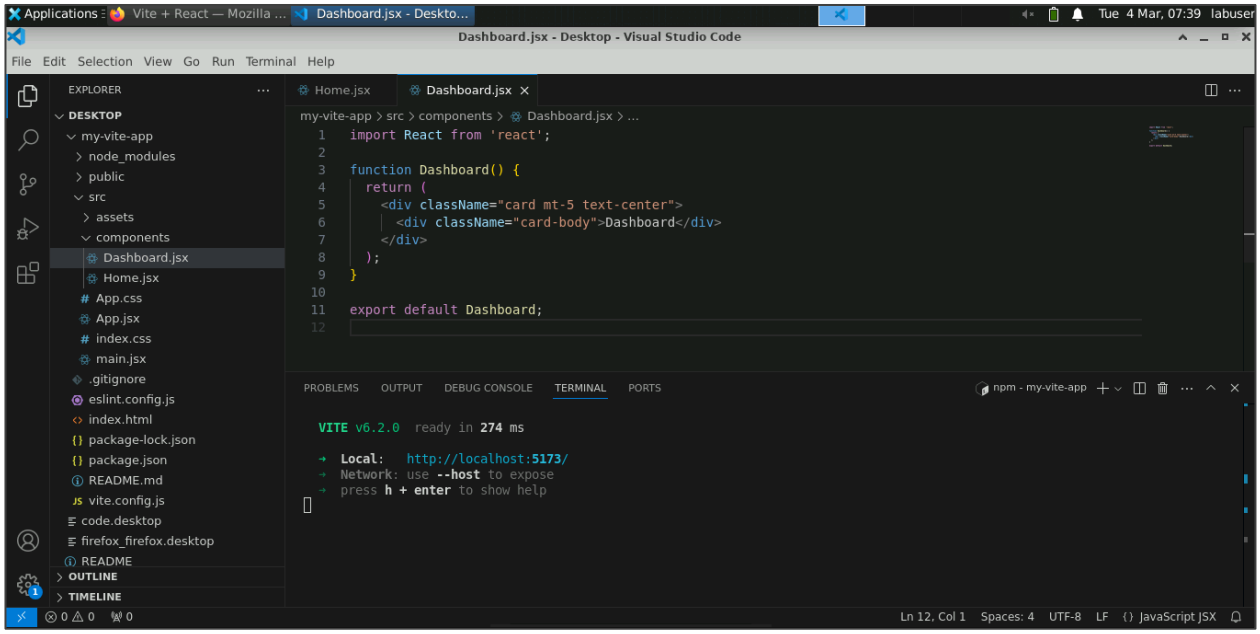


2.4 Inside the **components** folder, create a new file named **Dashboard.jsx**

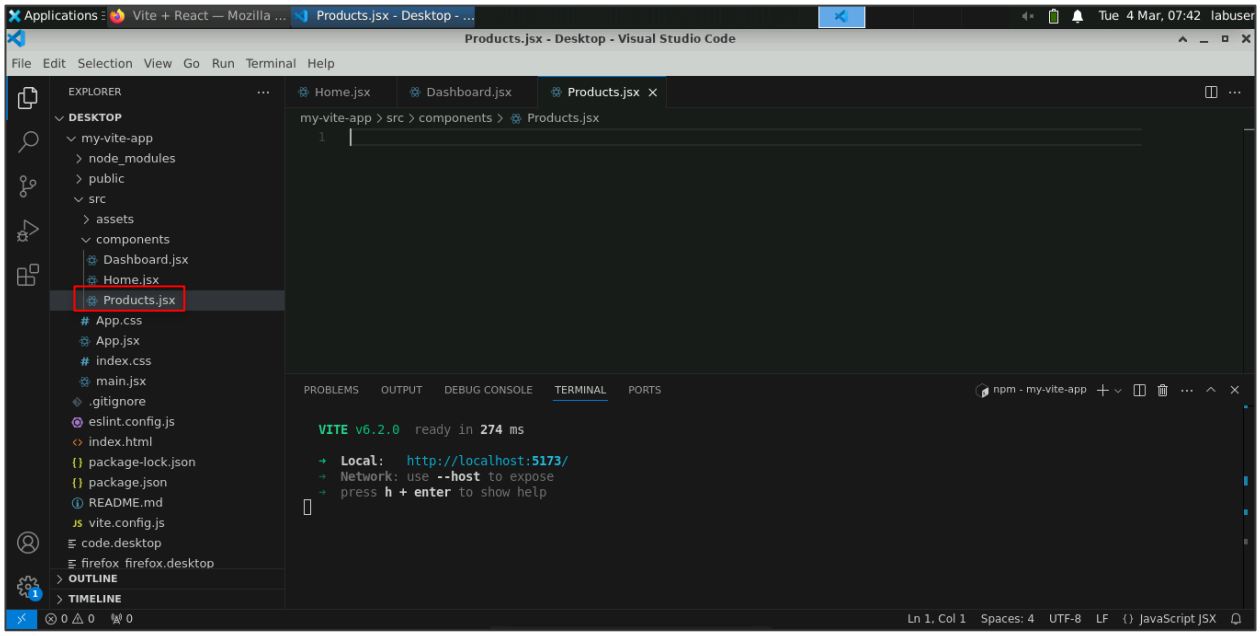


2.5 Add the following code to **Dashboard.jsx:**

```
import React from 'react';

function Dashboard() {
  return (
    <div className="card mt-5 text-center">
      <div className="card-body">Dashboard</div>
    </div>
  );
}

export default Dashboard;
```

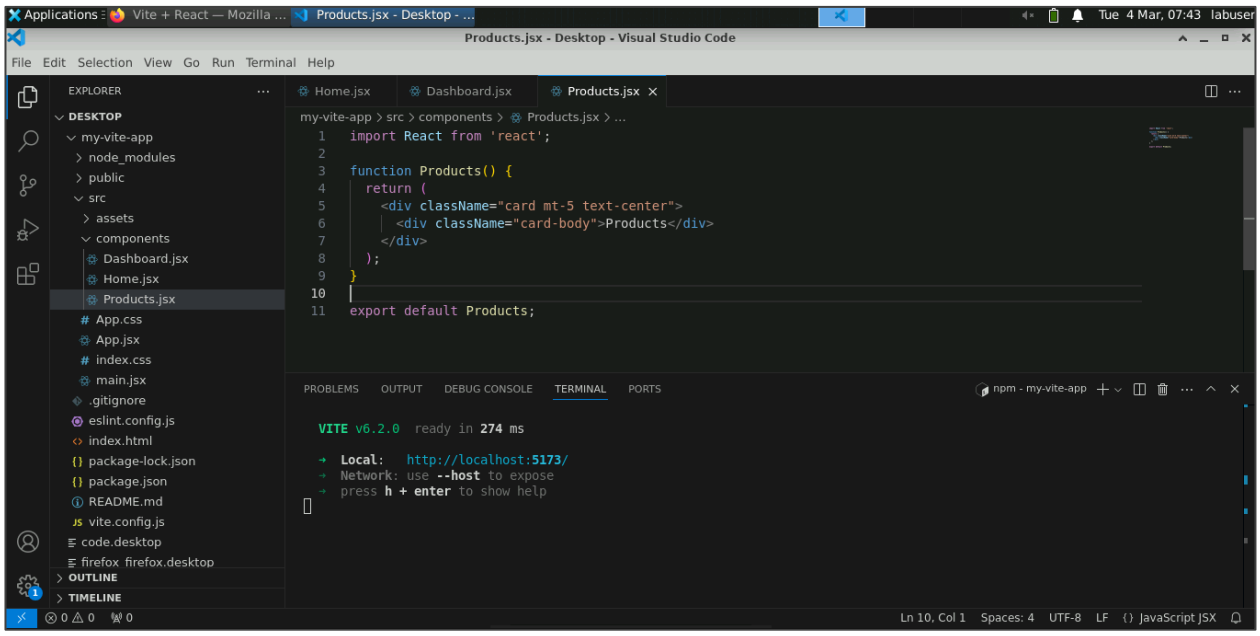


2.6 Inside the **components** folder, create a new file named **Products.jsx**

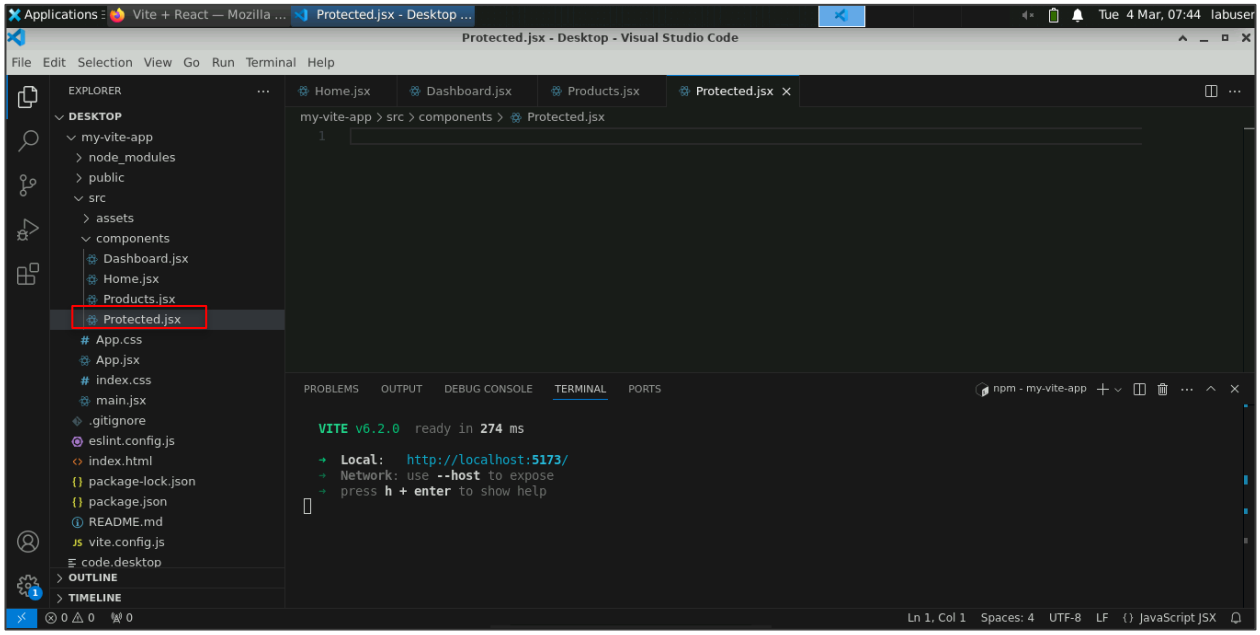


2.7 Add the following code to **Products.jsx:**

```
import React from 'react';

function Products() {
  return (
    <div className="card mt-5 text-center">
      <div className="card-body">Products</div>
    </div>
  );
}

export default Products;
```

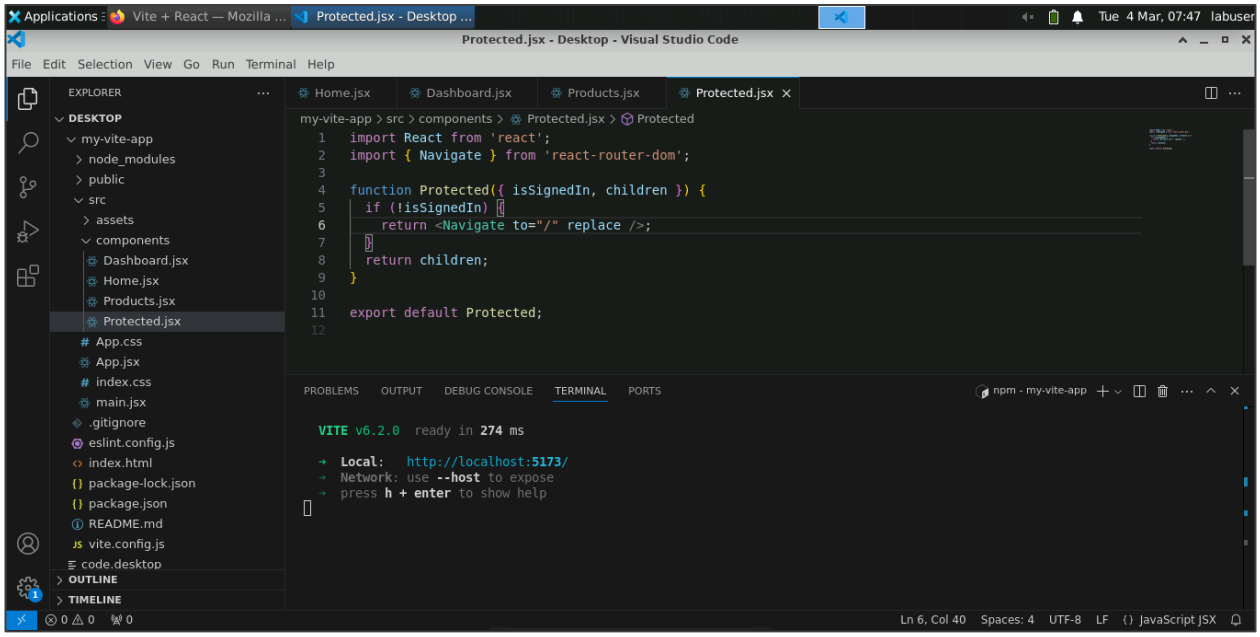


2.8 Inside the **components** folder, create a new file named **Protected.jsx**

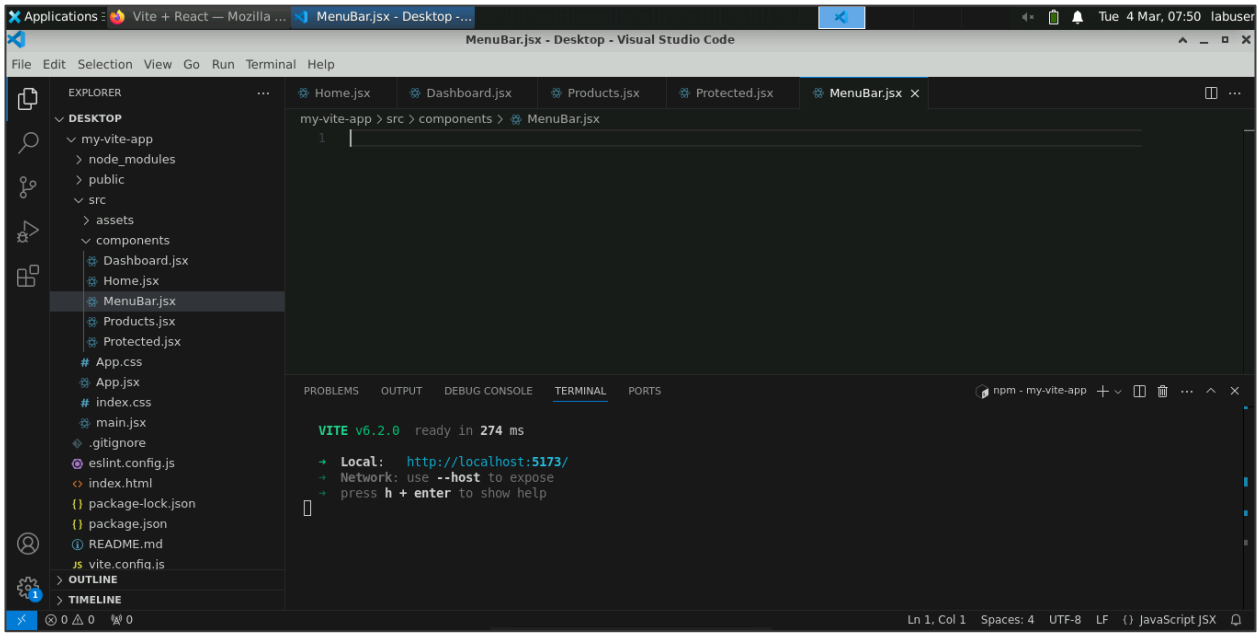


2.9 Add the following code to **Protected.jsx:**

```
import React from 'react';
import { Navigate } from 'react-router-dom';

function Protected({ isSignedIn, children }) {
  if (!isSignedIn) {
    return <Navigate to="/" replace />;
  }
  return children;
}

export default Protected;
```

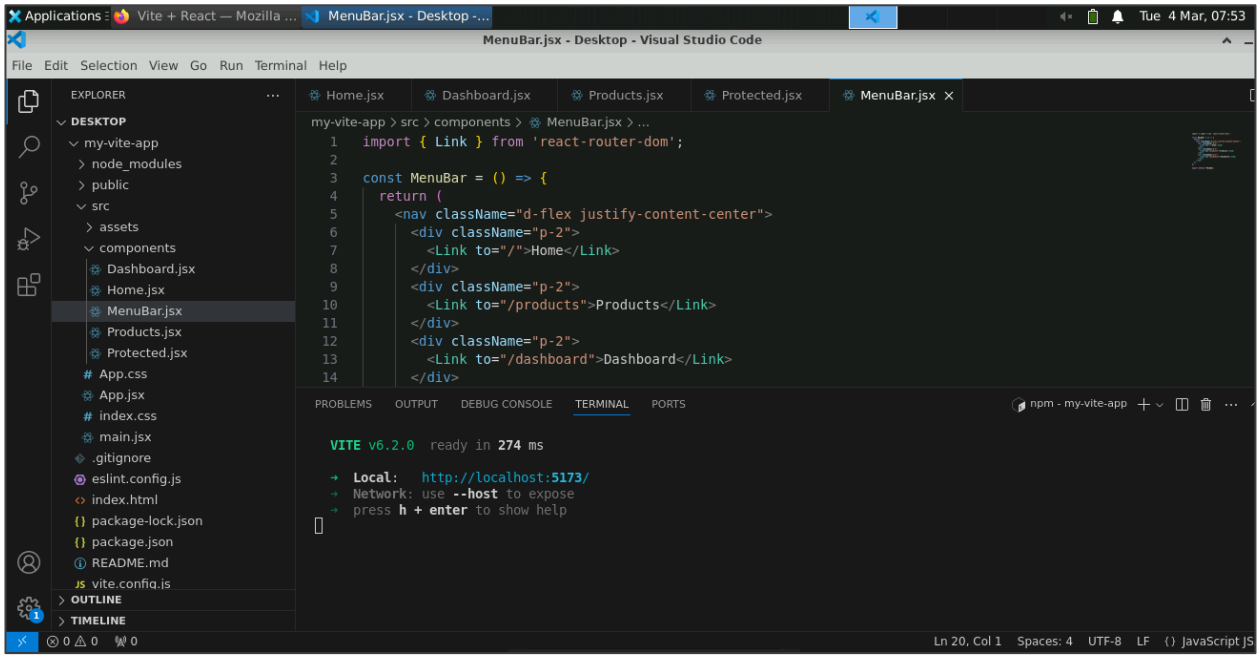


2.10 Inside the **components** folder, create a new file named **MenuBar.jsx**



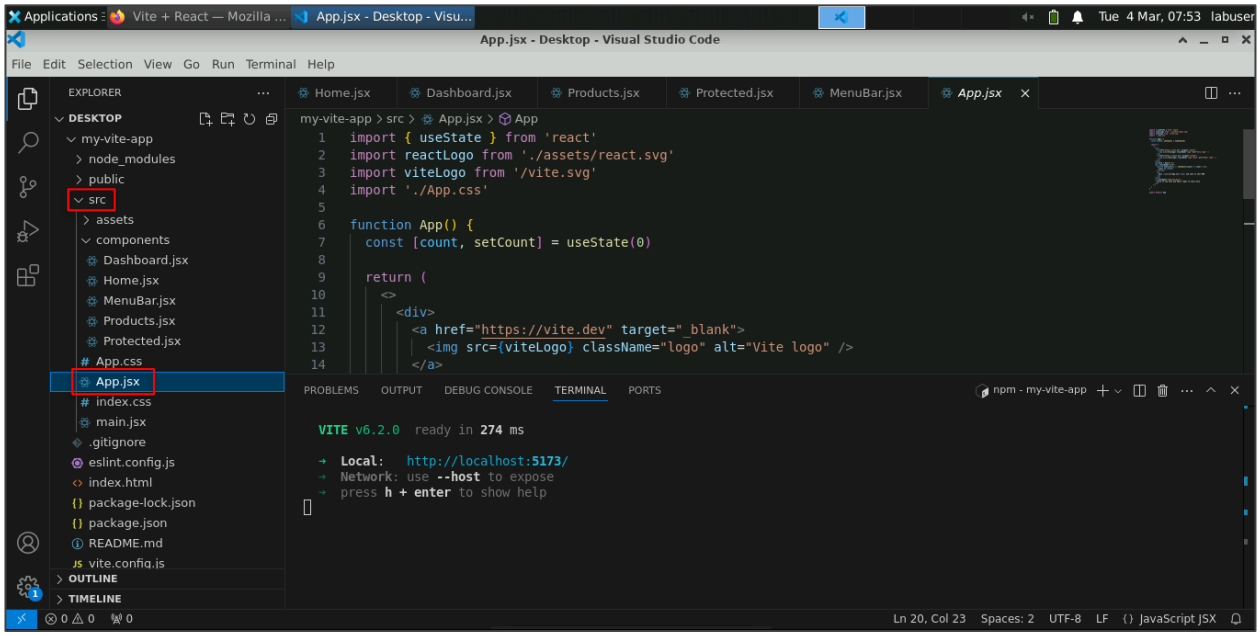
2.11 Add the following code to **MenuBar.jsx:**

```
import { Link } from 'react-router-dom';

const MenuBar = () => {
  return (
    <nav className="d-flex justify-content-center">
      <div className="p-2">
        <Link to="/">Home</Link>
      </div>
      <div className="p-2">
        <Link to="/products">Products</Link>
      </div>
      <div className="p-2">
        <Link to="/dashboard">Dashboard</Link>
      </div>
    </nav>
  );
};

export default MenuBar;
```



Step 3: Import and use the components

3.1 Open the **App.jsx** file inside the **src** folder

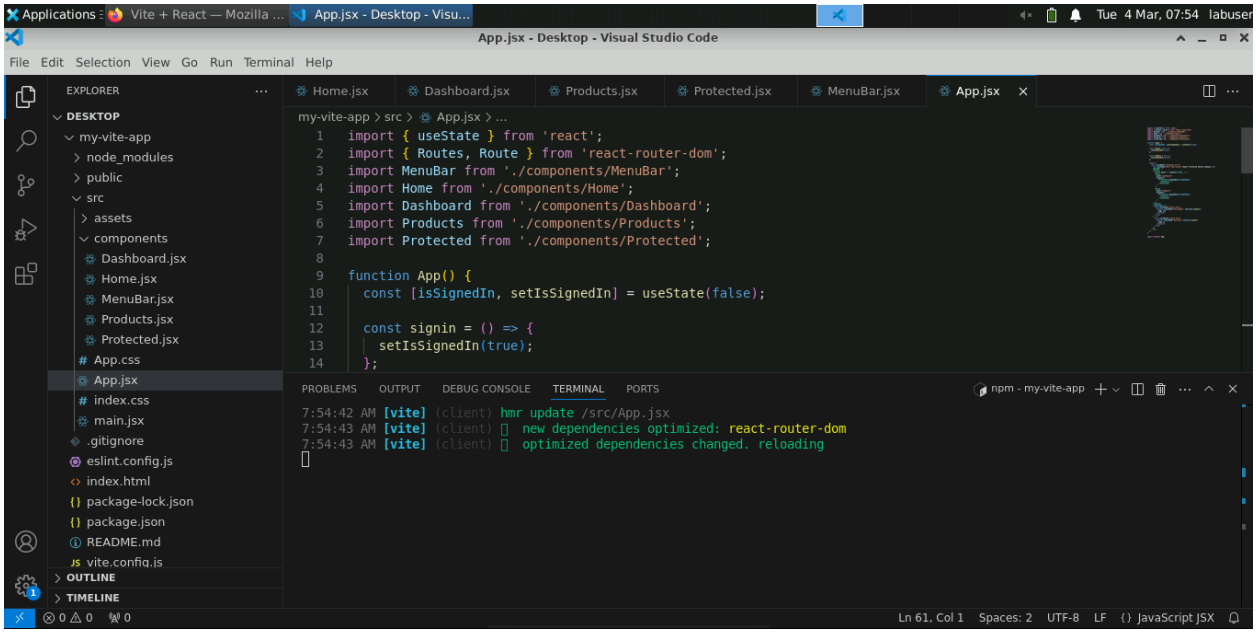


3.2 Paste the code below to include the new components in **App.jsx**:

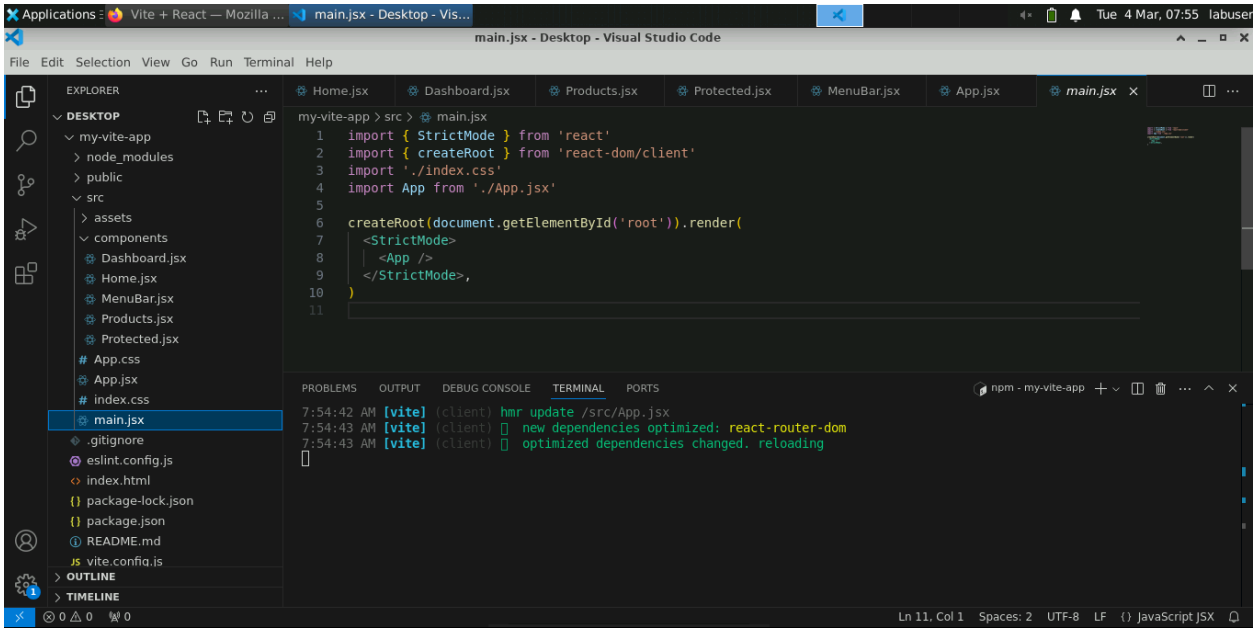
```
import { useState } from 'react';
import { Routes, Route } from 'react-router-dom';
import MenuBar from './components/MenuBar';
import Home from './components/Home';
import Dashboard from './components/Dashboard';
import Products from './components/Products';
import Protected from './components/Protected';

function App() {
  const [isSignedIn, setIsSignedIn] = useState(false);

  const signin = () => {
    setIsSignedIn(true);
  };

  const signout = () => {
```

```
  setIsSignedIn(false);
};

return (
  <div className="container mt-2">
    <h2 className="mb-5 text-center">React Protected Routes Example</h2>
    <MenuBar />
    <Routes>
      <Route path="/" element={<Home />} />
      <Route
        path="/dashboard"
        element={
          <Protected isSignedIn={isSignedIn}>
            <Dashboard />
          </Protected>
        }
      />
      <Route
        path="/products"
        element={
          <Protected isSignedIn={isSignedIn}>
            <Products />
          </Protected>
        }
      />
    </Routes>
    {isSignedIn ? (
      <div className="d-grid mt-2">
        <button className="btn-danger" onClick={signout}>
          Sign out
        </button>
      </div>
    ) : (
      <div className="d-grid mt-2">
        <button className="btn-dark" onClick={signin}>
          Sign in
        </button>
      </div>
```

```
    )}
  </div>
 );
}

export default App;
```

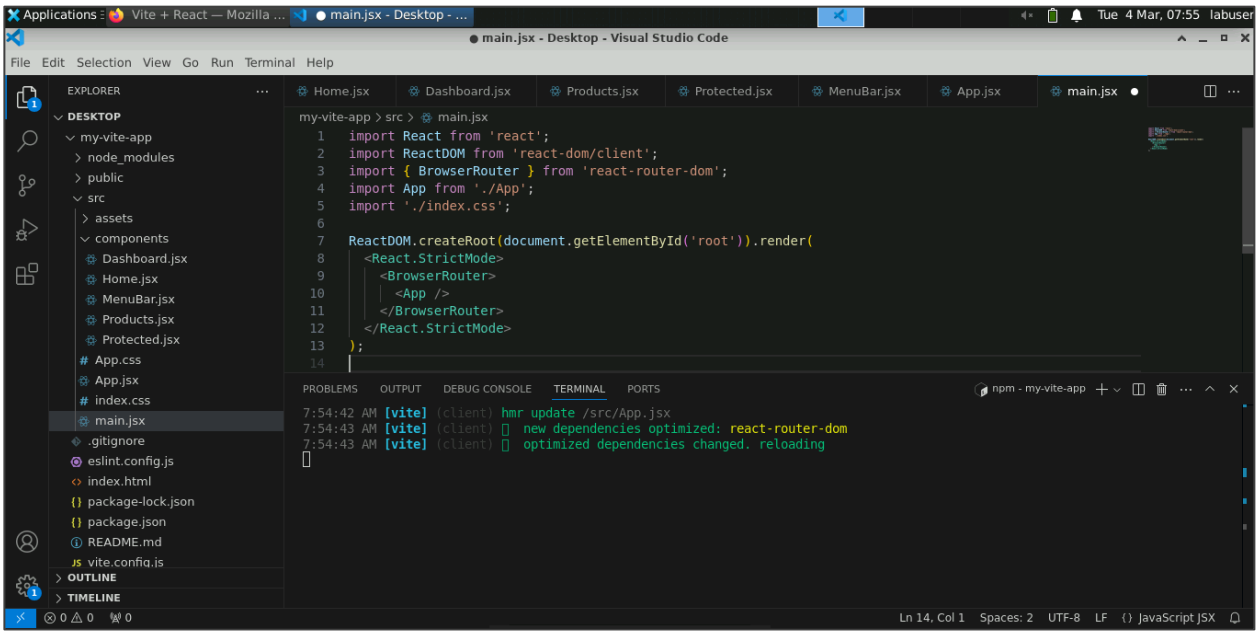


3.3 Open the **main.jsx** file inside the **src** folder



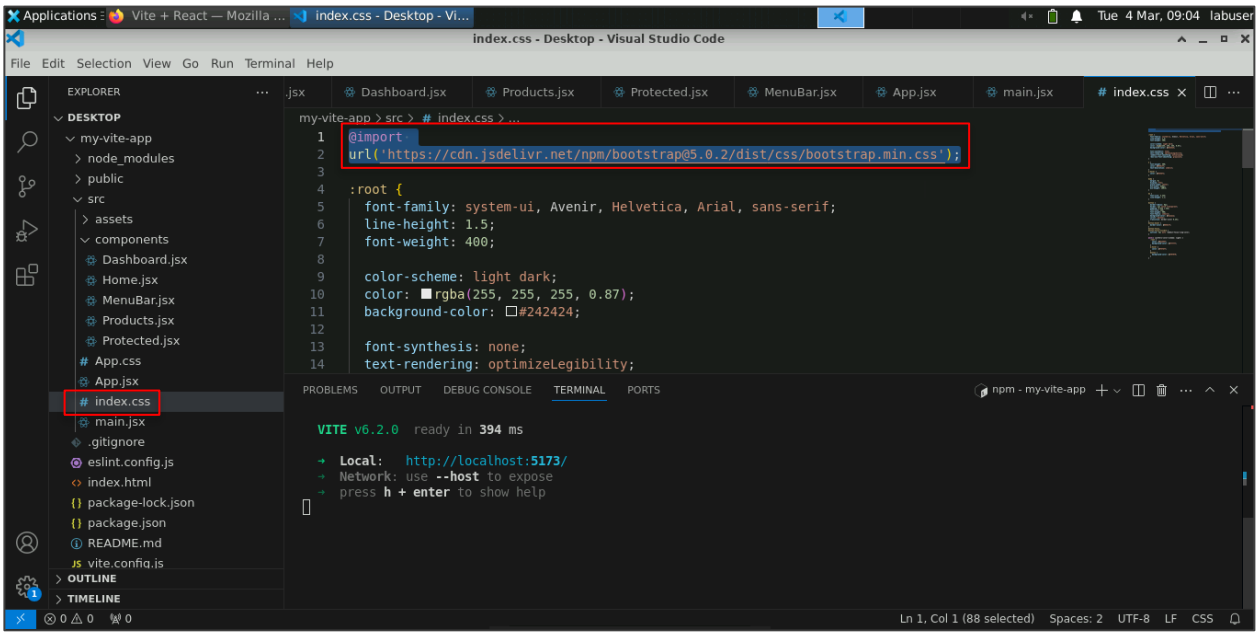
3.4 Paste the code below to include the new components in **main.jsx**:

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import { BrowserRouter } from 'react-router-dom';
import App from './App';
import './index.css';

ReactDOM.createRoot(document.getElementById('root')).render(
  <BrowserRouter>
    <App />
  </BrowserRouter>
);
```



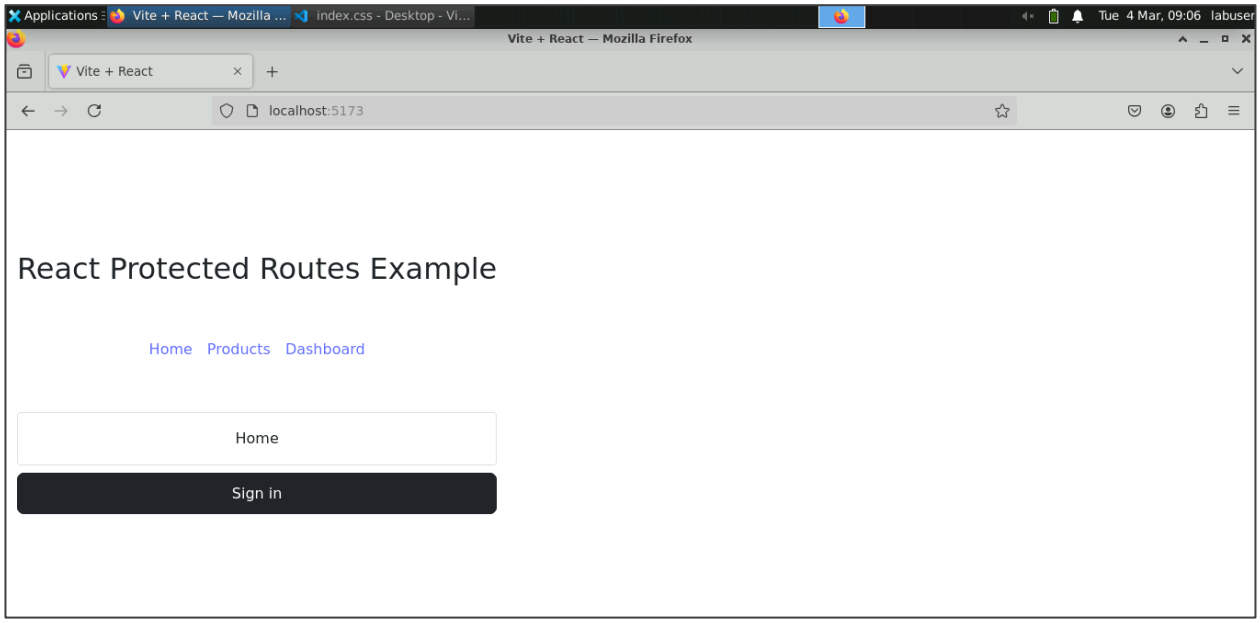
3.5 Open **index.css** file in **src** folder and paste the code below:

@import
url('https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/bootstrap.min.css');

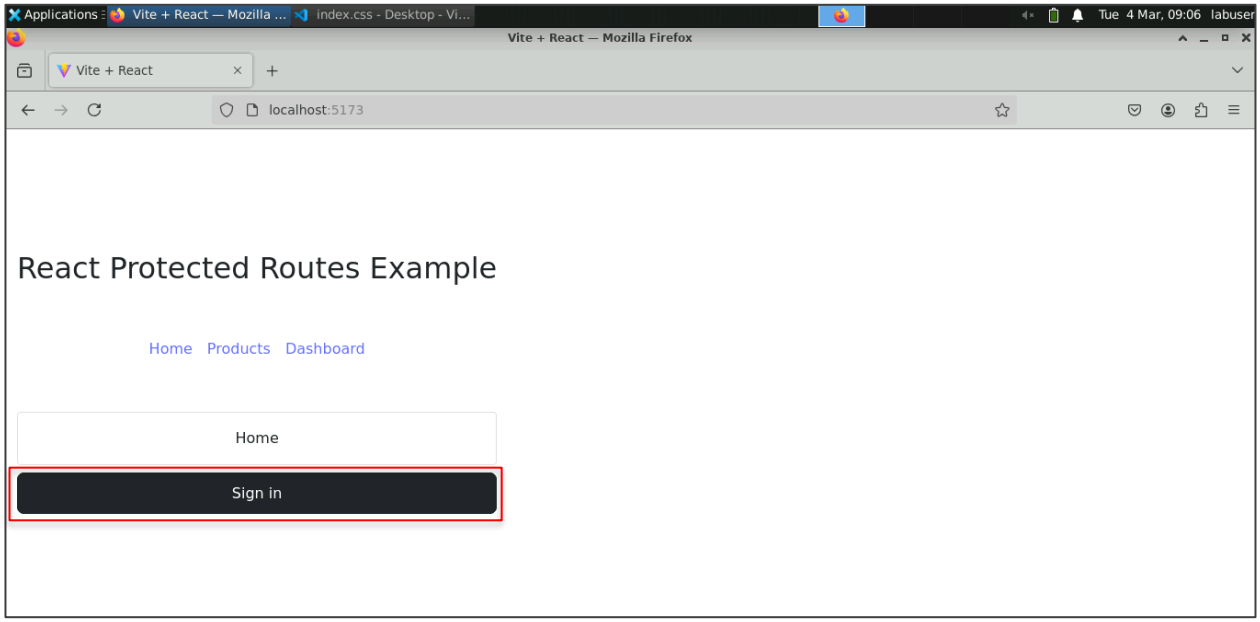


Step 4: Run and verify the application

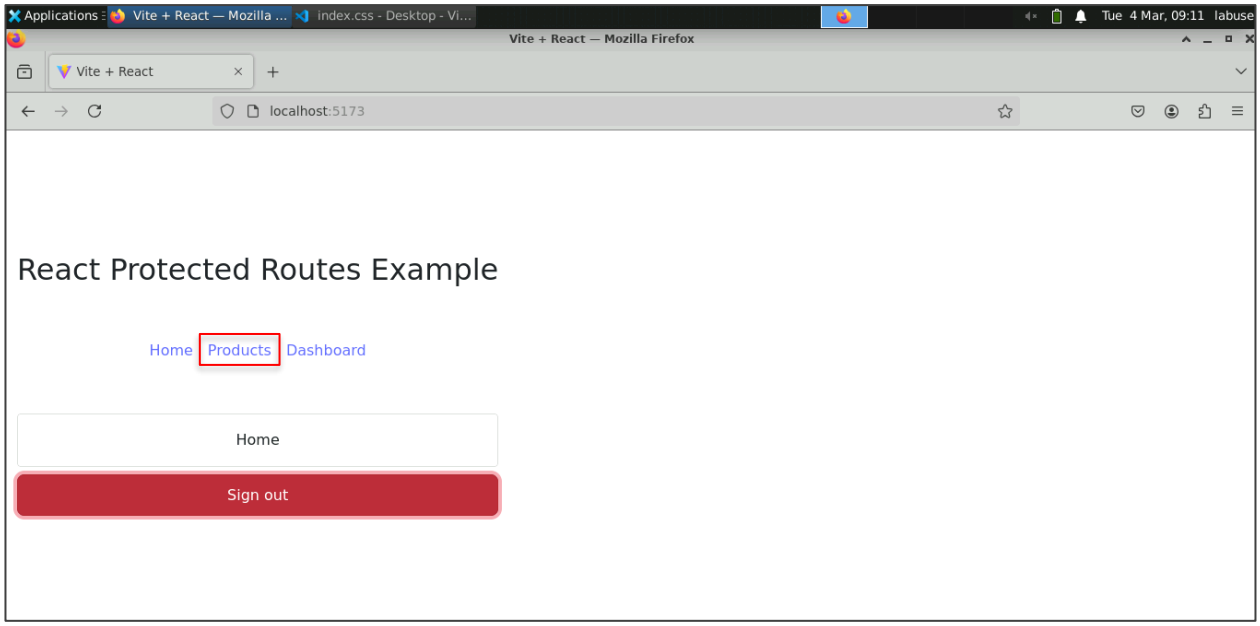
4.1 Navigate to the browser and refresh the webpage



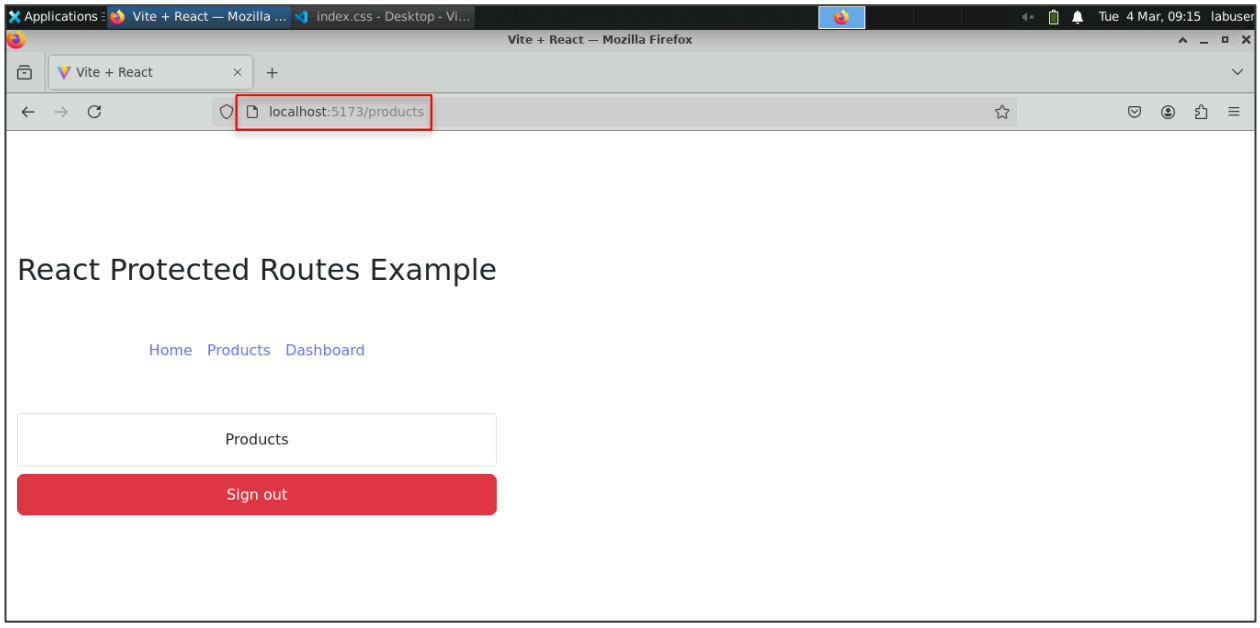
4.2 Click **Sign in** to enable navigation in the application



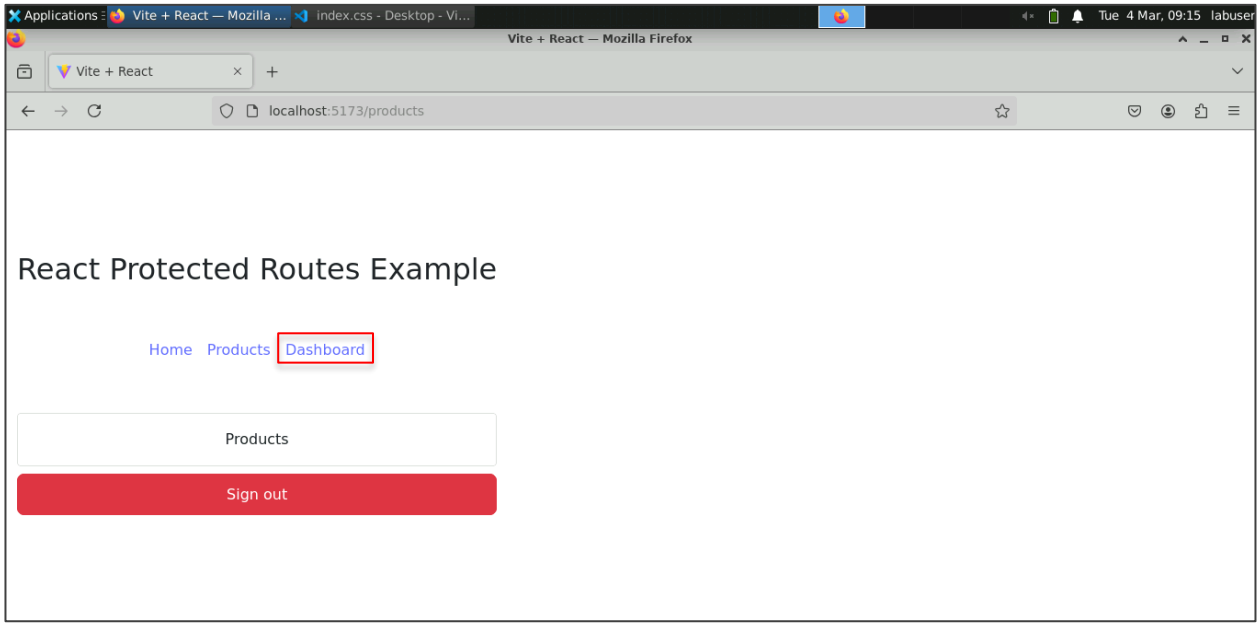
4.3 Click **Products** to navigate to **/products** route

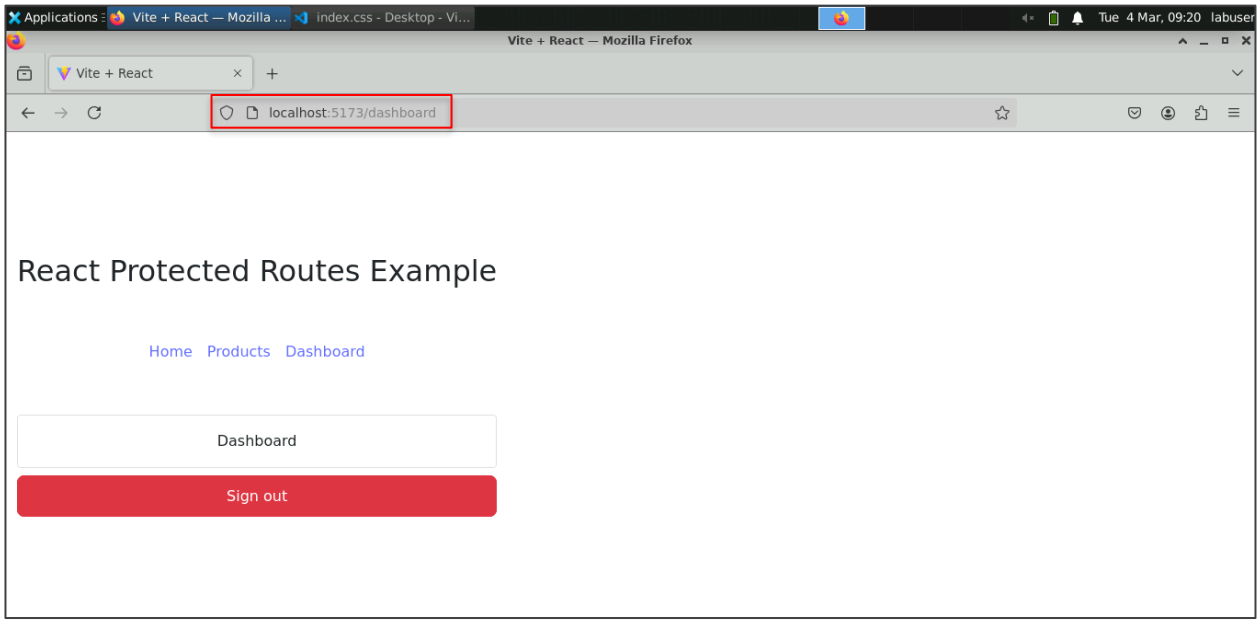


Note: This functionality will not work if the user is not signed in.



4.4 Click **Dashboard** to navigate to **/dashboard** route





By following the above steps, you have successfully implemented protected routes in a Vite-based React app, restricting access to specific pages based on user authentication. This enhances security and provides a foundation for future authentication features.